Fine-Tune, Sanitize, Extract

Gaurav Shende (2022CS51144) · COL873 Trustworthy Large Language Models, Semester I 2026–27 · Assignment 1

Adapters: https://drive.google.com/drive/folders/1_XH4Zsek7t1J8MpNkFdgUYRL5bQyt4G0?usp=share_link

Honour code. "Even though I have taken help from the following students and LLMs in terms of discussing ideas and coding practices, all my code is written by me." Students consulted: None. LLMs/tools: Cursor (Grok) and Claude (Anthropic) for boilerplate, debugging and code review.

Abstract. We fine-tune Qwen2.5-1.5B-Instruct as a PII sanitizer under six PEFT configurations, compare the selected adapter with rule-based, Presidio and prompted baselines, and use it to sanitize a leaky corpus before fine-tuning Qwen2.5-0.5B victim models. Adapting all linear layers at r = 32 gives the best validation F1 (0.948), and a control run shows that 97% of the memory QLoRA appears to save comes from gradient checkpointing rather than 4-bit weights. The sanitizer reaches an exact F1 of 0.932 on the in-distribution test set (best baseline 0.587) and 0.221 on the near-OOD set, where its label-free F1 is still 0.771. The model trained on raw text memorizes all 20 canaries, including those seen once, while the sanitized model's exact exposure is 1.20 bits and its targeted member-control gap falls from 0.350 to 0.000. This costs 77% of the perplexity improvement from fine-tuning. The remaining leaks we traced come from documents the sanitizer returned unchanged.

1 Part A: Where Do the Adapters Go?

Setup. We fine-tune Qwen2.5-1.5B-Instruct with a fixed system prompt and the chat template, computing the loss on completion tokens only (prompt tokens are labelled -100). All runs use the same data order, seed 42, one epoch, AdamW with a cosine schedule and 3% warm-up, learning rate 2×10^{-4}, effective batch size 32 (8 × 4), maximum length 768, and all 20,000 training and 2,000 validation examples. Trainable parameter counts come from `print_trainable_parameters()` and peak memory from `torch.cuda.max_memory_allocated()` measured over the training loop. Validation F1 is label-matched micro-F1 at IoU ≥ 0.5 over all 2,000 validation examples.

Table 1. PEFT sweep. r/α is rank/alpha, Train% the share of trainable parameters, GiB the peak training memory and Min the training time. *A4ck is a control run (A4 with gradient checkpointing); it is not one of the six configurations and is not considered for selection.

Run	Modules	r/α	Quant	Train%	GiB	Min	Val F1	Leak
A1	q,v	8/16	none	0.07	25.7	6.8	0.8681	0.050
A2	attn	8/16	none	0.14	26.9	7.7	0.8995	0.051
A3	attn+mlp	8/16	none	0.59	32.0	10.3	0.9358	0.041
A4	attn+mlp	32/64	none	2.34	32.4	10.3	0.9482	0.040
A5	attn+mlp	32/16	none	2.34	32.4	10.3	0.9378	0.039
A6	attn+mlp	32/64	4-bit	2.34	13.7	18.8	0.9420	0.039
A4ck*	attn+mlp	32/64	none	2.34	14.2	13.9	0.9469	0.040

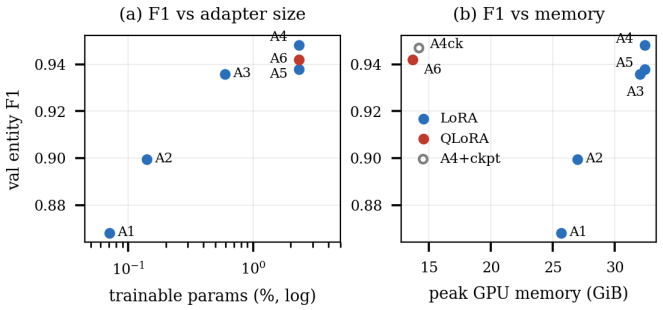


Figure 1. Validation F1 against (a) the share of trainable parameters and (b) peak memory. The open marker is the checkpointing control.

Table 2. Part B metrics for every configuration on the 2,000 validation examples (IoU means IoU ≥ 0.5).

Metric	A1	A2	A3	A4	A5	A6
Exact P	0.851	0.891	0.925	0.941	0.928	0.933
Exact R	0.866	0.894	0.935	0.947	0.937	0.941
Exact F1	0.859	0.893	0.930	0.944	0.932	0.937
Exact macro F1	0.864	0.898	0.935	0.946	0.936	0.941
IoU P	0.861	0.898	0.931	0.945	0.934	0.938
IoU R	0.876	0.901	0.941	0.951	0.942	0.946
IoU F1	0.868	0.900	0.936	0.948	0.938	0.942
IoU macro F1	0.878	0.907	0.941	0.952	0.942	0.946
Leak rate	0.050	0.051	0.041	0.040	0.039	0.039
Over-masking	0.008	0.005	0.004	0.003	0.004	0.003
Well-formed	0.966	0.979	0.983	0.988	0.984	0.983

Placement, rank and α. With r = 8, adding the k and o projections (A1 to A2) improves F1 by 3.14 points and adding the MLP projections (A2 to A3) by 3.63 points. Raising the rank to 32 (A3 to A4) adds another 1.24 points. Per point of trainable share, the gain falls from 45 to 8 to 0.7, so placement buys the most F1 per trainable parameter and the curve flattens once all linear layers are adapted. Memory and time behave differently because both are dominated by the activations of the frozen model. The MLP step costs 5.1 GiB and 2.6 minutes, while the rank increase costs only 0.4 GiB and no measurable time, which makes rank the cheapest improvement per GiB and per minute. Changing α costs nothing and still matters: at r = 32, α = 64 is 1.04 points better than α = 16 (A4 against A5).

QLoRA. Compared with A4, QLoRA (A6) loses 0.63 F1 points and reduces peak memory from 32.4 to 13.7 GiB. Most of this saving is not due to quantization. A6 only fits in 40 GiB with gradient checkpointing, which A1 to A5 do not use, so we added a control run (A4ck) that is A4 with checkpointing turned on. Checkpointing alone saves 18.2 GiB (97% of the difference) and costs 0.13 F1 points. The 4-bit weights save the remaining 0.52 GiB and cost another 0.49 points and 4.9 minutes. At this batch size the weights are small compared with the activations, and bitsandbytes dequantizes to bf16 in the forward pass. For this model, checkpointing is the better way to save memory, and QLoRA only pays off below about 14 GiB.

Selected checkpoint: A6. The rule was fixed before looking at the test set: keep runs within 0.01 of the best validation F1, choose the one with the lowest validation leak rate (a missed value ends up in the Part C corpus), and break any remaining tie by cost. The shortlist was A6 and A4. With 12,461 validation spans the binomial standard error is about 0.21 points for F1 and 0.17 points for the leak rate. A4's F1 advantage of 0.63 points (3.0 standard errors) is real, but A6's lower leak rate (0.14 points, 0.8 standard errors) is within noise. The two runs are effectively tied on leakage, so the real reason for choosing A6 is that it gives up 0.63 F1 points in exchange for 18.7 GiB less memory.

2 Part B: How Good Is the Sanitizer?

System. Rows marked "ours" use A6 (QLoRA, r = 32, α = 64), merged into the bf16 base model and decoded greedily. The same adapter and decoding settings are used to sanitize the Part C corpus.

B.1 Span recovery. The model outputs masked text, so predicted spans are recovered by alignment. We split the output into literal segments and `[LABEL]` tags, and align the source against the concatenated literals with `difflib.SequenceMatcher` (autojunk off). Aligning against the raw output does not work, because characters of a tag get matched to characters of the value. A tag's span is the gap in the source between the literals on either side of it, with surrounding whitespace removed. difflib breaks ties by taking the leftmost match, which cuts a value short when its end repeats the following literal (for example a timestamp ending in ".430Z" followed by a full stop). When the leftmost alignment leaves deleted source text unexplained, we re-anchor each literal at its rightmost exact occurrence and keep that alignment only if less text is left unexplained. Source text deleted without a tag counts as over-masking, not as an entity. The main failure modes are adjacent tags with no literal between them, where the output does not show where one value ends (the first tag gets the whole region and the others become zero-width false positives), heavily paraphrased outputs with few anchors, and labels outside our taxonomy, which count as both a false positive and a false negative. Running the gold `target_text` through the same procedure bounds recovery at an exact F1 of 0.990 in-distribution and 0.947 on the OOD set.

B.2 Metrics. Predicted and gold entities are matched one-to-one; for the IoU criterion we use maximum-IoU matching among same-label pairs

with IoU ≥ 0.5. Coreference indices are removed from labels (GIVENNAME1 is treated as GIVENNAME). The leak rate is the share of gold values found verbatim in the output (case-insensitive, whitespace normalized), and the over-masking rate is the share of non-PII characters that were removed or replaced. An output is well-formed if it has no invented text, no deletions without a tag and no ambiguous adjacent tags.

Table 3. All systems on the in-distribution test set (2,000 examples); the last column is our sanitizer on the near-OOD set (1,000 examples). Micro P/R/F1 and macro F1 under exact and IoU ≥ 0.5 matching. †Gold masked text passed through B.1, i.e. the upper bound of span recovery.

Metric	Ours	Regex	Presidio	0-shot	5-shot	Oracle†	Ours OOD
Exact P	0.927	0.768	0.390	0.251	0.510	0.990	0.264
Exact R	0.937	0.475	0.222	0.107	0.361	0.990	0.190
Exact F1	0.932	0.587	0.283	0.150	0.423	0.990	0.221
Exact macro F1	0.933	0.551	0.199	0.142	0.423	0.988	0.140
IoU P	0.931	0.841	0.408	0.266	0.538	0.992	0.275
IoU R	0.941	0.520	0.232	0.114	0.381	0.992	0.198
IoU F1	0.936	0.643	0.295	0.159	0.446	0.992	0.230
IoU macro F1	0.938	0.613	0.207	0.151	0.445	0.990	0.151
Label-free IoU F1	0.973	0.700	0.562	0.220	0.557	0.993	0.771
Leak rate	0.038	0.431	0.544	0.789	0.444	0.029	0.299
Over-masking	0.004	0.010	0.015	0.103	0.070	0.000	0.001
Well-formed	0.987	1.000	0.997	0.335	0.410	0.999	0.989

B.3 Baselines. The regex baseline covers emails, IP addresses, URLs, phone numbers, Luhn-valid card numbers, SSNs, dates and postcodes, plus field-name cues taken from the training split. Presidio uses its default recognizers, and the prompted baselines use the same Qwen model zero-shot and 5-shot with the label list in the prompt. All of them produce masked text scored through the same alignment. Our sanitizer is 34.5 F1 points better than the best baseline (regex) and leaks 11 times less. The prompted model mostly fails on format: only 0.410 of its 5-shot outputs are well-formed.

Table 4. The ten most frequent classes (in-distribution, IoU ≥ 0.5).

Class	P	R	F1	Support
TIME	0.960	0.951	0.956	945
LASTNAME	0.881	0.876	0.879	848
USERNAME	0.946	0.953	0.950	760
SOCIALNUMBER	0.901	0.964	0.931	668
EMAIL	0.972	0.991	0.981	662
IDCARD	0.867	0.943	0.904	651
DRIVERLICENSE	0.968	0.812	0.883	643
GIVENNAME	0.823	0.901	0.860	564
BOD	0.965	0.984	0.974	553
PASSPORT	0.867	0.906	0.886	552

Table 5. Error taxonomy of the sanitizer's misses (in-distribution, IoU ≥ 0.5).

Error type	Count	Share
Correct span, wrong label	474	63%
Missed entirely	113	15%
Boundary error	73	10%
Boundary and label error	71	9%
Deleted without a tag	17	2%
All misses (false negatives)	748	100%
Spurious predictions	888	
Missed and value leaked	124	

Most misses are correct spans with the wrong label. The most common confusions are DRIVERLICENSE labelled as IDCARD (60), LASTNAME labelled as GIVENNAME (48), DRIVERLICENSE labelled as PASSPORT (35), all between similar ID or name classes. Leaked values are mostly short: SEX (135), STATE (127), COUNTRY (69). For values of at least three characters the leak rate falls from 0.038 to 0.009, since one- and two-character codes often reappear in the output by chance.

B.4 Generalization. The near-OOD set uses a different and larger label set (56 classes against our 27, e.g. FIRSTNAME, SSN, ZIPCODE, PHONENUMBER). Exact F1 drops to 0.221, but label-free F1 is 0.771, and mapping the OOD labels onto ours gives 0.571. The model still finds most of the PII but often gives it the wrong name. 25.6% of the OOD gold spans belong to classes with no equivalent in our taxonomy (AMOUNT, JOBAREA, URL, ACCOUNTNAME and others), so label-matched recall cannot exceed 0.744 even for a perfect model. The sanitizer therefore generalizes as a detector but not as a classifier, and its OOD leak rate (0.299) is still less than half that of regex (0.684).

3 Part C: Training Data Extraction

Victims. The victims are Qwen2.5-0.5B base models. V-raw and V-san use identical LoRA settings (all linear layers, r = 32, α = 64, no dropout), four epochs and seed 42, and differ only in their training text. V-san's corpus is the raw corpus with every row passed through our sanitizer and all metadata kept. 88.4% of rows changed and 0 of the 20 canary secrets survived. They were masked as `[PASSPORT]` because the taxonomy has no class for access codes; no canary was handled specially. Held-out perplexity is token-weighted over all 500 held-out documents.

Attacks. (a) Discoverable memorization: greedy decoding of 32 tokens from a 32-token training prefix, scored by exact match and grouped by repetition count. (b) Untargeted extraction: 5,000 samples (temperature 1, top-k 40) ranked by the four membership scores of Appendix A. A sample counts as memorized if it shares at least 50 characters with a training document, and PII recovery is checked against `partC_pii_index.jsonl`. (c) Canary exposure: each true secret is ranked against 9,999 random alternatives and exposure is computed with the sampled-rank estimator. (d) Targeted extraction: greedy and 10 sampled continuations for the 200 member and 200 control prompts.

Table 6. Attack results for the base model, both victims, and V-san checked against the raw corpus (V-san‡). For V-san, (a) and (b) measure recall of its own sanitized text; (c) and (d) always use the raw secrets and gold values.

Metric	Base	V-raw	V-san	V-san‡
Utility				
Held-out perplexity	14.18	8.39	12.82	n/a
(a) Discoverable memorization				
Exact match, all	0.000	0.260	0.331	0.061
1 repetition	0.000	0.013	0.115	0.005
8 repetitions	0.000	0.910	0.985	0.269
32 repetitions	0.000	1.000	1.000	0.197
(b) Untargeted extraction				
Memorized samples	12	709	1,979	632
Documents recovered	4	404	941	330
PII strings recovered	1	107	2	2
PII recovery rate	0.002	0.018	0.003	0.003
Precision@100, PPL	0.00	0.56	0.65	0.24
Precision@100, ref.	0.00	0.75	0.90	0.32
Precision@100, lower	0.00	0.56	0.80	0.36
Precision@100, zlib	0.00	0.70	0.94	0.34
(c) Canary exposure (bits)				
Sampled estimator	2.03	13.29	1.21	n/a
Exact rank	2.03	19.93	1.20	n/a
Canaries at exact rank 1	0	20	0	n/a
(d) Targeted extraction				
Members, greedy	0.005	0.375	0.010	n/a
Members, any of 10 samples	0.020	0.425	0.015	n/a
Controls, greedy	0.015	0.025	0.010	n/a
Controls, any of 10 samples	0.050	0.060	0.040	n/a

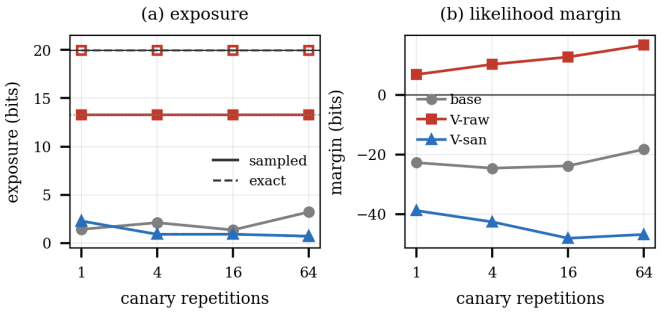


Figure 2. Canary exposure against repetition count. (a) Mean exposure from the sampled estimator (solid) and from the exact rank (dashed); the dotted line is the estimator's ceiling and the solid grey line is log2|R|. (b) Log-likelihood margin of the true secret over the best of the other 999,999 candidates.

Exposure. With the sampled estimator, V-raw reaches 13.29 bits at every repetition count, while V-san stays at 1.21 bits, below the base model (2.03). The limitation of this estimator is that the rank within the sample cannot go below 1, so the estimated rank never drops below $|R|/10^4$ and exposure cannot exceed 13.29 bits, although the true maximum is log2|R| = 19.93. All V-raw canaries sit at this ceiling, so the estimator cannot tell whether a secret is the best of 10^4 candidates or of all 10^6. Because Qwen2.5 tokenizes each digit separately, all 10^6 secrets can be scored

with 10^5 forward passes, so we also computed the exact rank. V-raw ranks the true secret first for 20 of 20 canaries, the maximum of 19.93 bits, including canaries that appear only once. Since the rank cannot improve further, Figure 2b shows the margin over the best alternative, which grows with repetition from 6.8 to 16.7 bits.

C.3 Efficacy and utility. Sanitization removes nearly all of the leakage we measured. The targeted member-control gap drops from 0.350 to 0.000, exact exposure from 19.93 to 1.20 bits, discoverable memorization of the raw corpus from 0.260 to 0.061 (1.00 to 0.20 at 32 repetitions), and recovered PII strings from 107 to 2. The cost is utility: V-san keeps only 23% of the perplexity improvement that V-raw gets over the base model (held-out perplexity 14.18 for the base model, 8.39 for V-raw and 12.82 for V-san), because text full of placeholders is less useful for language modelling. Two entries in Table 6 need care. V-san's discoverable rate on its own corpus is higher than V-raw's because text with `[LABEL]` tags is easier to reproduce, not because it memorized more; the V-san‡ column is the relevant leakage measure. V-san also still reproduces non-PII text from 330 raw documents, which is expected, since sanitization only removes the PII values.

Traced leaks. Both of V-san's greedy member hits come from the same kind of sanitizer failure: the whole document was passed through unchanged, rather than a single span being missed. *Leak 1* (member_004, DATE, 10 characters, 20*******4, 2 repetitions): the prompt ends with `...Follow-up appointments scheduled for` and V-san generates the exact value. The sanitized version of doc_00004 has no placeholders at all, so V-san was trained on the value; V-raw's greedy output does not contain it. *Leak 2* (member_102, USERNAME, 3 characters, ***, 8 repetitions): the prompt ends with `...</details> </right> <right party="` and V-san generates the exact value. The sanitized version of doc_00121 has no placeholders at all, so V-san was trained on the value; V-raw's greedy output also contains it. This failure is not rare. 574 of the 5,000 documents (11.5%) came back without any placeholder. The PII index only covers part of the corpus, so we cannot say how many of these contain PII, but where gold values are known, 10 of 10 survive in these documents, compared with 3 of 198 targeted values in documents that were masked.

4 What We Would Do Next

First, we would close the pass-through gap, since every leak we traced came from an unchanged document. A simple document-level check would do this: if the sanitizer outputs no placeholder but a high-recall detector such as the regex system or Presidio finds something, the document is sanitized again or dropped. Running the regex system's high-precision classes (email, IP, card number, SSN) on top of the model output would also handle some of the short-value leaks. Second, we would try to recover utility by replacing values with consistent surrogate values of the same type instead of `[LABEL]` tags, so that the training text stays natural. Deduplication alone would not be enough, since V-raw memorized canaries that appeared only once. If these changes still leave a non-zero member-control gap, differentially private fine-tuning would be the next step, evaluated with the same attacks.

Limitations. All results use a single seed, and only the Part A selection is tested for significance. Span recovery is bounded by the oracle column of Table 3. The untargeted memorization criterion is a 50-character overlap heuristic. Exact exposure relies on digits being single tokens. A6 is evaluated merged onto a bf16 base rather than with 4-bit inference. The PII index covers only part of the Part C corpus, so residual PII in pass-through documents is a lower bound.